

Covalent bonding

COVALENT BONDING IS WHEN ATOMS FORM BONDS BY SHARING ELECTRONS.

Rather than giving away or accepting electrons, some atoms share their outer electrons to achieve full outer shells.

Sharing electrons

Covalent bonds are formed of pairs of electrons, one from each atom. The pair is included in the outer electron shell of both atoms at once. This allows the atoms to have a full set of eight electrons in their outer shell and become stable. No electrons leave their original atoms—so the atoms always remain neutral.

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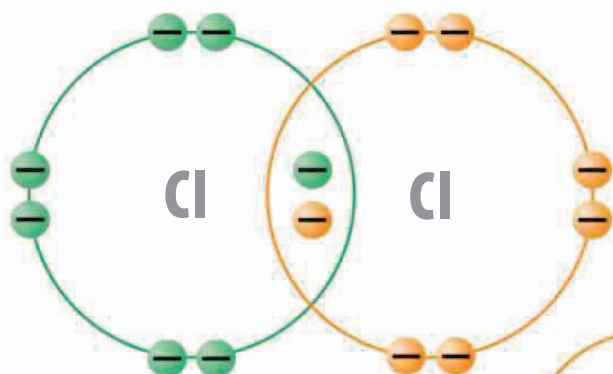
◀ 110–111 Compounds and molecules

◀ 112–113 Ionic bonding

Hydrogen bonds 142 ▶

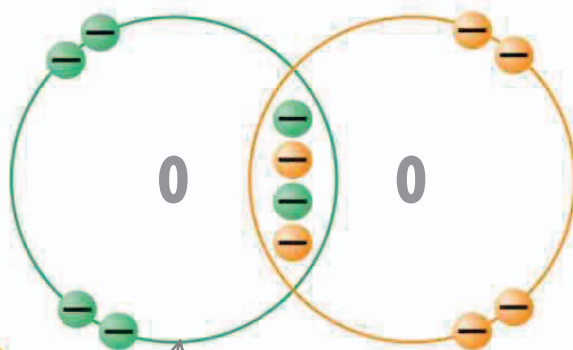
Hydrocarbon chains 158 ▶

Inside atoms 168–169 ▶



△ Single bond

Chlorine (Cl) has one space in its outer shell. In a chlorine molecule (Cl_2), two chlorine atoms are bonded by a single pair of shared electrons.



▽ Double bond

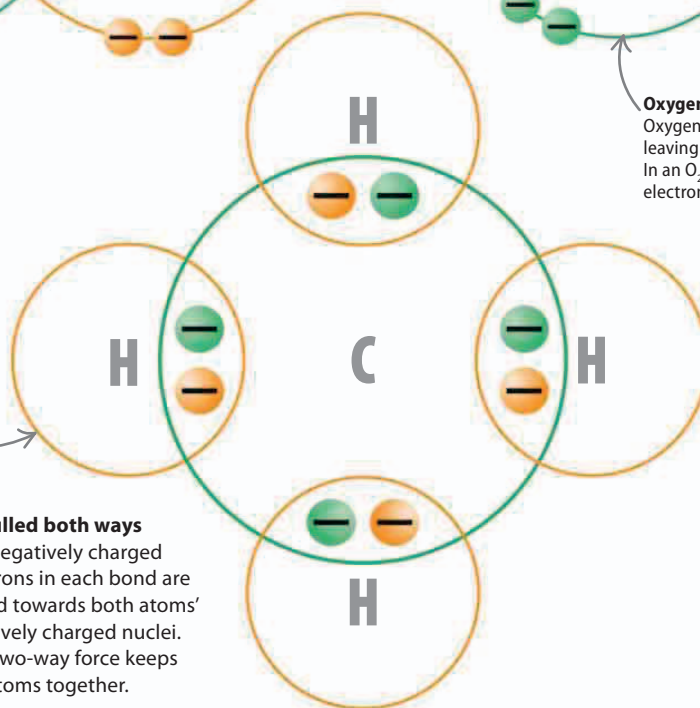
Oxygen gas is made up of O_2 molecules. They form when oxygen atoms share not one but two pairs of electrons in what is known as a double bond.

Oxygen molecule

Oxygen (O) atoms have six outer electrons, leaving two spaces in their outer shells. In an O_2 molecule, each atom shares two electrons with its neighbor.

Methane molecule

Carbon (C) has four spaces in its outer electron shell and can form four covalent bonds at once. It bonds with hydrogen atoms to form methane (CH_4).



△ Pulled both ways

The negatively charged electrons in each bond are pulled towards both atoms' positively charged nuclei. This two-way force keeps the atoms together.

Diamond, the hardest substance of all, is held together with just covalent bonds. However, most covalently bonded substances are soft and brittle.